# Uniform Stability of the Spherical Laughlin Gap:
Physical Gram Transfer, Improved Charge Exponents, and Quantified Relative Bounds

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**Abstract**

We examine the full spherical fermionic $V_1$ interaction at flux $q = 3(N-1)$ with coefficient one for every pair projector, and bounded real scalar disorder projected to the lowest Landau level. The source claims uniform stability for all sufficiently large $N$, with existential constants, using an imported Fock-space gap and additional retained-block, pinning, charge, and local-loss estimates. We preserve that attribution. Physical-metric and residual tools from the private Ram corpus are applied to the actual redundant four-body wedge maps, giving an explicit pair-energy error bound without inverting a singular Gram matrix. We extend the source's slow-charge step to every fixed exponent below $1/3$ and optimize its annular split: the choice $1/4$ followed by $3/2$ gives a conditional $O(L^{-1/10})$ centered-interaction budget in place of the displayed $O(L^{-1/24})$ budget. We also quantify the dissipative relative-bound closure and the moving-ground-state stability argument, including the necessary exact annihilation and number-sector conditions. Finite exact arithmetic tests are separated from uniform analytic proofs; the public Lean scope is recorded without claiming an independent kernel build. No new unconditional many-body gap theorem or practical disorder threshold is claimed. Contribution assessment: C3/5, notable source-specific restricted refinements with inherited global analytic inputs.

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# 1 Problem, Quantifiers, and Status

The pinned manuscript is OpenAI, *Uniform Stability of the Spherical Laughlin Gap*, October 5, 2026, study manuscript M545, family 269 [1]. The study uses the preserved initial 722 manuscript IDs; these IDs are not a claim about a later live repository count. The applicability row assigns L4. That relevance classification is independent of the realized C3 contribution assessment here.

Let $U_q$ be the spin-$q/2$ lowest Landau level on the round sphere of radius $\sqrt{q/2}$, area $2\pi q$. Its orthonormal homogeneous-polynomial basis is $e_j = \sqrt{\binom{q}{j}} z_0^{q-j} z_1^j$, $0 \le j \le q$. All spaces below are complex; inner products are conjugate-linear in the first argument. Define

$$\mathcal{F}_q = \bigoplus_{n=0}^{q+1} \bigwedge^n U_q, \qquad H_q = \bigoplus_n H_{n,q}, \qquad H_{n,q} = \sum_{i<j} P_{ij}^{(1)}.$$

Here $P_{ij}^{(1)}$ is the pair projection onto spin $q-1$; it is not a truncation to a thin cylinder. Every operator and quadratic form at fixed $q$ has domain the entire finite-dimensional space. A uniform statement still quantifies over dimensions tending to infinity. For real measurable $\varphi \in L^\infty(S_q)$ put

$$T_q(\varphi) = \Pi_q M_\varphi \Pi_q|_{U_q}, \qquad V_q(\varphi) = d\Gamma(T_q(\varphi)).$$

The source's main claim is

$$\begin{aligned} \exists \lambda_*, \Delta_* > 0,\ N_* \quad \forall N \ge N_*\ \forall \varphi = \overline{\varphi},\ \|\varphi\|_\infty \le 1\ \forall |s| \le \lambda_* : \\ E_1(s) - E_0(s) \ge \Delta_*. \end{aligned} \tag{1}$$

The eigenvalues are those of $H_{N,3(N-1)} + sV_q(\varphi)$, counted with multiplicity. The potential may depend on $N$; it need not be radial, smooth, or probabilistic. The ground line moves with $s$.

The imported theorem in source Section 2 is

$$\begin{aligned} \forall \gamma \in (0, \gamma_*)\ \exists q_\gamma\ \forall q \ge q_\gamma : \quad H_q^2 \ge \gamma H_q \text{ on } \mathcal{F}_q, \\ \gamma_* = \frac{4616733319001}{10^{14}} > \frac{1}{25}. \end{aligned} \tag{2}$$

It is attributed to the companion Fock-space manuscript [2]. The present report does not replace its proof by small-system spectra. The retained-block estimate, stronger than (2), must also be supplied. The restricted results below either prove finite identities directly or state the source estimates on which they depend.

## 1.1 Current public verification scope

On October 8, 2026 the primary public notes `lean/docs/269.md` describe unperturbed spherical, Fock, and planar gap formalizations, with comparator links `LaughlinGap.lean`, `Laughlin.lean`, `LaughlinFock.lean`, and `LaughlinPlanar.lean`. They explicitly exclude projected-disorder stability and uniqueness of its perturbed ground state from that selected scope [3]. This is a public assertion about formalization scope, not an independently executed Lean kernel verification by this report. The inspected comparator targets contain `sorry` placeholders; they are statement interfaces, not implementation proofs checked here. The spherical target uses antisymmetric configuration functions and distance from the polynomial Laughlin line, with bounds 1/25 and 1/100 in the two spherical files. The Fock target explicitly builds occupation-basis CAR matrices and quantifies over every $0 < \gamma < \gamma_*$ before choosing a flux threshold. Its pair coefficient converts the tensor normalization by a factor $\sqrt{2}$, matching increasing wedges. No claim of absent formalization is made. The full pinned mathematical source was read, including Sections 1–8, the retained-block appendix, bibliography, and its seven data rows; bundled drawing-library sources are not proof evidence.

# 2 Source Chain and Ram Inputs

| Source location | Logical role and required uniformity |
|---|---|
| Section 2 | Exact pair normalization, zero-mode algebra, imported Fock gap. |
| Section 3 | Tight localized frame, CAR locality, polynomial propagation, time filters, common flux padding; constants independent of $q, n$. |
| Section 4 | Rounded annihilator-column comparison, pinning gap, affine-subtracted transport and a one-sided negative-square bound. |
| Section 5 | Orthogonal flag histories, uniformly local polar map, centered readout, one-hole Schur average. |
| Section 6 | Slow and summable pooled charge; hole-linear local expectation bound on all zero spaces, not only a special quasihole family. |
| Appendix, Section 7 | Exact finite Gram comparisons, relative finite-flux transfer, retained four-body energy, local particle-loss channel. |
| Section 8 | Fixed-locality-norm relative-bound bootstrap and spectral transport of the perturbed ground line. |

No one line implies all later lines. In particular, a Fock gap is not the retained-block inequality, and a finite Gram test is not a locality theorem.

## 2.1 Actual private proofs used

Three full local files were read, rather than treating routing maps as proofs. Volume 21, `ch01d-physical-metrics.tex`, proves the physical Gram formula, metric symmetry, and the exact residual decomposition. For injective synthesis $J$, $G = J^*J$, $P = JG^{-1}J^*$, and $B = G^{-1}J^*AJ$, its proof gives

$$\|(A - z)Jc\|^2 = \|(B - z)c\|_G^2 + \|(I - P)AJc\|^2. \tag{3}$$

The two terms are the orthogonal components of the residual. The omitted component vanishes only under actual invariance, not because a compression was formed. Its metric-comparison proof also requires relative, not merely entrywise, Gram control. We adapt this account to a *noninjective* wedge synthesis below; the adaptation uses the quotient by its kernel, not an unjustified $G^{-1}$.

TR-item-047-astra, section `sec:soserror047`, proves that Gram negativity and an identity defect must be subtracted from a claimed lower bound. In its notation the subtraction is $\epsilon_e + \sum_i \eta_i B_i G_i$. We transfer that elementary quadratic-form argument to actual pair-generated annihilator columns: the error is paid in $H_q$, not in a volume-sized identity. This is a use of an existing residual principle, not a claim that SOS or matrix positivity is new.

Volume 21, `ch01f-spectral-limits.tex`, proves a nested graph-core pollution example and a two-sided approximate-inverse criterion. Its diagonal example is indefinite and is *not* asserted to model this positive Hamiltonian. Its relevant lesson is the quantifier distinction: fixed-vector convergence does not control moving vectors or all $q$. Our finite-flux transfer supplies an actual norm estimate on a fixed local Fock factor and then tensors with spectators. Exact paths, literal read ranges, and SHA-256 hashes are in the confidential local read ledger.

## 2.2 Relation to primary literature

The current primary literature checks confirm established localized-frame CAR locality [4], frustration-free stability with local conditions [5], bulk-gap stability in a GNS setting [6], and spectral flow along gapped paths [7]. These are not interchangeable models. None is used here as an automatic theorem for the full expanding spherical LLL interaction with arbitrary bounded scalar symbols. The source builds the relevant locality, charge, and loss inputs. The contribution here is the explicit source-specific

transfer and exponent/constant refinements, not the invention of frames, Schur averaging, or spectral flow.

## 3 Exact Pair Algebra and Physical Coordinates

For an increasing wedge $a$, let $B(a)$ be the adjoint of left exterior multiplication by $a$. Thus $B(a\wedge b) = B(b)B(a)$. Define the positive lift by $\mathcal{L}_k(|a\rangle\langle b|) = B(a)^*B(b)$. An orthonormal basis for the pair subspace is

$$v_p(i,j) = (j-i)\sqrt{\frac{\binom{q}{i}\binom{q}{j}}{q\binom{2q-2}{p}}}\,\mathbf{1}_{i+j=p+1}, \quad i<j, \qquad H_q = \sum_{p=0}^{2q-2} B(v_p)^*B(v_p). \tag{4}$$

To verify normalization, set $D = x\partial_x$ and $f = (1+x)^q$. Then $fD^2f - (Df)^2 = qx(1+x)^{2q-2}$, whose coefficient at $x^{p+1}$ is $\sum_{i<j}(i-j)^2\binom{q}{i}\binom{q}{j}$. Different $p$ have disjoint index sums, hence are orthogonal. The lift on $\bigwedge^n U_q$ has exactly coefficient one per pair, since the normalized contraction by a pair is $B/\sqrt{\binom{n}{2}}$.

Let $W_3(v\otimes u) = v\wedge u$ and $W_4(v\otimes w) = v\wedge w$, with the domain of $W_4$ using *ordered* pair factors. Normal ordering gives the full-Fock identity

$$H_q^2 = H_q + \mathcal{L}_3\{W_3(W_3^*W_3 - I)W_3^*\} + \mathcal{L}_4(W_4W_4^*). \tag{5}$$

Indeed zero inner contractions give the ordered sum $\sum_{p,r} B(v_p\wedge v_r)^*B(v_p\wedge v_r)$; two contractions give $H_q$ since the two-particle Hamiltonian is a projection. On three particles $H_q = W_3W_3^*$, fixing the remaining normal coefficient as $(W_3W_3^*)^2 - W_3W_3^*$. These coefficients determine the normal-ordered polynomial on every sector. No factor two may be added to the last term.

**Proposition 3.1** (Redundant Gram test)**.** *For any finite synthesis $W$ and Hermitian $C$, with $G = W^*W$,*

$$GCG \succeq 0 \quad\Longleftrightarrow\quad WCW^* \succeq 0.$$

*If $GCG \succeq -\epsilon G^2$, then $WCW^* \succeq -\epsilon WW^*$.*

*Proof.* Both inequalities test $C$ on $\operatorname{ran} W^* = \operatorname{ran} G$. For the second, replace $C$ by $C+\epsilon I$ and use the first. This is the physical quotient of the redundant coefficient space; no inverse or numerical rank choice is involved. □

For example, $W = (1\ \ 1)$ and $C = \operatorname{diag}(1,-1)$ give $WCW^* = 0$ although $C$ is indefinite. Demanding $C \succeq 0$ loses a valid certificate. Conversely $GCG \succeq -\epsilon I$ alone needs a lower bound on the positive spectrum of $G$ to recover a physical error. As $G = \operatorname{diag}(1,t^2)$ approaches singularity, such conversion is not uniform.

## 4 Retained Blocks and Quantitative Finite-Flux Transfer

The source appendix changes the common phase of the pair rows to give $v_0 = -e_0\wedge e_1$. We use that phase throughout this section; all Hamiltonians are unchanged. For fixed $p$,

$$v_p^*(i,j) = (i-j)\sqrt{\frac{p!}{2^p i!\,j!}}\,\mathbf{1}_{i+j=p+1}.$$

The star here denotes a limiting vector, not an adjoint. For $1 \le D \le 23$, the copy labels are the odd integers $r \in \mathcal{R}_D = [1, D] \cap (2\mathbb{Z}+1)$. The actual highest wedge columns and their limiting Gram matrices are

$$w^q_{Dr} = \sum_{\substack{p+j+k=D \\ j<k}} S^q_r(D, D; p, j, k) v_p \wedge e_j \wedge e_k, \qquad G_D = (\langle w^*_{Dr}, w^*_{Ds} \rangle)_{rs}.$$

The source's seven squares yield real symmetric copy matrices $E^q_D$ with limits $E_D$, and target copy matrix $(d_D/d_a)\chi$, where $d_D = 4q - 1 - 2D$, $d_a = 2q - 1$ and $\chi_{rs} = \mathbf{1}_{r=s=1}$. The limiting checks are

$$G_D(2\chi - E_D + \epsilon I)G_D \succeq 0, \qquad \epsilon = 3/10^6, \quad 1 \le D \le 23. \tag{6}$$

These finite comparisons consume only total-spin deficits $0, \dots, 22$. The blocks of deficits at least 23 remain in the square with coefficient one. Inverting $G_D$ would be wrong: these matrices have exact kernels.

**Theorem 4.1** (Explicit pair-energy transfer budget)**.** *Use the source's exact coefficient definitions above. On the local Fock factor of modes* $0, \dots, 24$*, let*

$$d_q(x) = \sqrt{\prod_{a=0}^{x-1}(1 - a/q)}, \qquad \Delta_q|A\rangle = \prod_{x \in A} d_q(x)|A\rangle, \qquad q \ge 24.$$

*For* $b = (p, j, k)$ *with* $p + j + k = D$*,* $j < k$*, set*

$$A_b = B(v^*_p \wedge e_j \wedge e_k), \quad \beta^q_{rb} = S^q_r(D, D; p, j, k)\frac{r_p(q)}{d_q(j)d_q(k)}, \quad r_p(q) = \sqrt{\frac{(2q)^p}{(2q-2)_p}}.$$

*Write*

$$C^q_D = (d_D/d_a)\chi - E^q_D + \epsilon I, \quad M^q_D = (\beta^q)^T C^q_D \beta^q, \quad M^*_D = (\beta^*)^T(2\chi - E_D + \epsilon I)\beta^*.$$

*Let* $m_D$ *be the largest multiplicity of a fixed* $p$ *among these* $b$*. Assume* (6)*, and supply rigorous bounds* $\|M^q_D - M^*_D\| \le e_D(q)$ *and* $\|M^*_D\| \le m^*_D$*. Then on the entire physical Fock space,*

$$\sum_{r,s}(C^q_D)_{rs} B(w^q_{Dr})^* B(w^q_{Ds}) \ \ge -\rho_D(q) H_q, \tag{7}$$

*where one valid explicit choice is*

$$\rho_D(q) = m_D\left[e_D(q)e^{2300/(q-23)} + m^*_D(e^{2300/(q-23)} - 1)\right]. \tag{8}$$

*In particular, verified* $e_D(q) \le K_D/q$ *give a uniform* $O(q^{-1})$ *transfer error for* $q \ge 46$*, with computable constants.*

*Proof.* Checking the removed labels in an increasing wedge gives the exact annihilator conjugations

$$B_p = r_p(q)\Delta_q^{-1} B(v^*_p)\Delta_q, \qquad B(v_p \wedge e_j \wedge e_k) = \frac{r_p(q)}{d_q(j)d_q(k)}\Delta_q^{-1} A_b \Delta_q.$$

The inverse diagonal acts on the output sector. Repeated labels give zero on both sides. Since $r_p \ge 1$ and $0 < \Delta_q \le I$,

$$\Delta_q h_* \Delta_q \le h_q, \quad h_* = \sum_{p=0}^{23} B(v^*_p)^* B(v^*_p), \quad h_q = \sum_{p=0}^{23} B^*_p B_p \le H_q.$$

Moreover, CAR contractions have norm at most one, so $\sum_b \|A_b\phi\|^2 \le m_D\langle\phi, h_*\phi\rangle$. After the two outer diagonals, the quadratic expression being estimated is $\sum_{b,c}(M_D^q)_{bc}A_b^*\Delta_q^{-2}A_c$. Its limiting expression $\sum_{b,c}(M_D^*)_{bc}A_b^*A_c$ is positive: Proposition 3.1 first gives positivity on four particles, and the positive lift gives positivity on local Fock space. This does not assert $M_D^* \succeq 0$ on every redundant coefficient vector.

The difference is the block operator $R = M_D^q \otimes \Delta_q^{-2} - M_D^* \otimes I$ on the column of annihilator outputs. Thus

$$\|R\| \le e_D(q)\|\Delta_q^{-2}\| + m_D^*\|\Delta_q^{-2} - I\|.$$

Its form is bounded by $m_D\|R\|\, h_*$, using Cauchy–Schwarz in that physical output space. Conjugating back gives the same coefficient times $h_q$. Finally, $-\log(1-t) \le t/(1-t)$ yields

$$\log\|\Delta_q^{-2}\| = \sum_{x=0}^{24}\sum_{a=0}^{x-1} -\log(1-a/q) \le \frac{\sum_{x=0}^{24} x(x-1)/2}{q-23} = \frac{2300}{q-23}.$$

The diagonal has eigenvalues at least one, so its distance from $I$ is its largest eigenvalue minus one. This proves (8). For $q \ge 46$, $q - 23 \ge q/2$, and $e^t - 1 \le te^t$ proves the rate. All operators are even and supported on these 25 modes. Tensoring the inequality with the identity on the remaining modes proves it for any number of spectator particles. □

The rate hypothesis $e_D(q) \le K_D/q$ is deliberately not inferred from sampled fluxes. It can be established by bounds on the finitely many falling-factorial and coupling formulas on a specified interval $q^{-1} \in [0, Q^{-1}]$. Entrywise certified errors give a Frobenius bound. The exponential bound is conservative, especially near $q = 24$; its role is a valid finite-flux account, not an economical threshold.

### 4.1 Paying for all finite comparisons

The source's orthonormal highest-vector coefficients obey Bessel's inequality before wedging:

$$\sum_{r\in\mathcal{R}_D} \|B(w_{Dr}^q)\psi\|^2 \le \sum_{\substack{p+j+k=D\\ j<k}} \|B_p\psi\|^2.$$

There are $n_D = \lfloor (D+1)^2/4 \rfloor$ triples. Schur averaging gives $d_a \int B_p(g)^* B_p(g)\, dg = H_q$. Consequently the total allowance in (6) is exactly bounded by $\epsilon\sum_{D=1}^{23} n_D H_q = 1222\epsilon H_q$. If the block transfer bounds are first made relative to the full $h_q$ as in the proof above, their averaged total is bounded by $24\sum_D \rho_D(q)H_q$. This is conservative and explicit.

For completeness the negative three-body tail is paid by

$$\delta_q = \sum_{\substack{17\le z\le q\\ z \text{ odd}}} \frac{3q-1-2z}{2q-1}(z+1)\frac{(q)_z}{(2q-2)_z}\left(3z-1-\frac{z(z+1)}{q}\right).$$

The pair-generated highest-vector Cauchy–Schwarz bound supplies the factor $z+1$. For $q \ge 4$ the falling-factorial ratio is at most $(2/3)^z$ and the dimension ratio is at most two. Hence summands are dominated by $6z(z+1)(2/3)^z$; dominated convergence is legitimate and $\delta_q \to 61/4096$. Fixed-$z$ convergence alone would not suffice.

The seven-square two-body coefficient is $\eta = 93527408868499/10^{14}$. If the finite-flux three-body inequalities hold and all stated transfer bounds are verified, the resulting retained coefficient is at least

$$c(q) = 1 - \eta - \delta_q - 1222\epsilon - 24\sum_{D=1}^{23}\rho_D(q). \tag{9}$$

Thus $c(q) > 0$ certifies $H_q^2 \ge c(q)H_q + \mathcal{L}_4(W_4 R_q^{\mathrm{hi}} W_4^*)$. We have not computed a rigorous usable $q$ threshold from this formula. The limiting arithmetic margin is exactly $1 - \eta - 61/4096 - 1222\epsilon = \gamma_*$.

# 5 Pinning and Charge: Assumptions That Cannot Be Skipped

Write $Z_{n,q} = \ker H_{n,q}$. Absence of pair relative angular momentum one forces divisibility of an antisymmetric polynomial by the cube of each pair bracket. Distinct brackets are irreducible and nonassociate; their product therefore divides the polynomial. The remaining factor is symmetric of individual spinor degree $h = q + 3 - 3n$. Consequently

$$Z_{n,q} = \{\text{Laughlin cubic product} \times S_h\}, \qquad \dim Z_{n,q} = \binom{n+h}{n} \quad (h \geq 0, n \geq 1).$$

For $h < 0$ the space vanishes. At $q = 3(N-1)$, the filled zero space is one-dimensional and no zero mode has $n > N$; the vacuum is treated separately. Contraction of the south orbital gives orthogonal branch coordinates $Z_{n,q} \simeq Z_{n,q-1} \oplus Z_{n-1,q-3}$. The first branch records a flag and decreases $h$ by one; the second removes an electron and preserves $h$. Flags are coordinates, not measured hole positions.

Multiplying by $d = 1$ or 3 pinned zeros introduces $g(j) = \frac{1}{2}\sum_{r=0}^{d-1} \log((q-j-r)/(q-r))$. This map is not unitary. The source rounds its south-pole singularity at fixed $B$ and compares columns of normalized pair annihilators. Its squared column error is $C\sum_b w_b^5/(v_b+B)^4 \leq CB^{-1/2}$. Equal nullities follow from the invertible kernel map $e^{sD(g)}$, so singular-value min–max transfers a gap. Removing the rounding costs a fixed local condition number $K_B$, giving a pinning gap of order $\gamma/(4C_B^2K_B^2)$, independent of particle number. A direct norm estimate on the extensive Hamiltonian would not give this.

Use $u = 1+m$, $v = 1+q-m$, $w^2 = uv/(q+1)$ and $f_{L,p}(m) = (L/(L+m))^p$. The transported weighted number observable has interaction size

$$A_{q,p}(m) = \frac{u^2 f''_{L,p}(m)}{(q+1)^2} \asymp_p \frac{f_{L,p}(m)}{(q+1)^2}\left(\frac{u}{L+u}\right)^2. \tag{10}$$

Affine subtraction is essential: a constant plus index-linear symbol commutes with the number and axial generators. The pinning rate is $w^2/v^2$; its product with $w^2 f''$ gives (10). Source Section 4 proves the one-sided zero-space bound $PK_{q,d}(f)P \leq C\int A_{q,p}/w\, dm\, P$. Its sign is obtained after a positive-frequency filter, not from individual row pieces. With raising pieces $F_x = -\rho_x G_x + E_x$, $\rho_x = f''/(-g'') > 0$, the leading compression is a negative square $-2P(\int \sqrt{\rho_x}G_x)^*(\int \sqrt{\rho_x}G_x)P$. Commutator errors cost $A/w^2$ and $A/w$; the latter is dominant.

## 5.1 Precisely retained analytic inputs

The next theorem assumes the source's pinning, local-history, and one-hole estimates, with constants uniform in $n, q, L$ at each fixed weight exponent. They are actual estimates in Sections 4–5, not consequences of finite matrix positivity. In history coordinates set

$$S_{L,p}(\omega) = \sum_{b\in\mathcal{B}(\omega)} f_{L,p}(b), \quad W_m(\omega) = \sum_b \min\{1, ((1+m)/(1+b))^a\}, \quad 1 < a < 2.$$

The assumed centered readout is $\pm(\mathcal{R}POP\mathcal{R}^* - \mathrm{ref}_q(O)I) \leq Cr^B W_m$ for neutral Hermitian radius-$r$ observables of norm one. For rapid tails it costs $\sum_\nu r_\nu^B \|O_\nu\|$. The reference is a positive linear functional, not an independently chosen scalar for every shell. In a one-quasihole space ($h = 1$), the assumed integrated comparison between any two normalized zero modes is $Cr^{B'}$, independent of area. This comes from irreducibility of that space, dimension $n+1$, and bounded trace of $W_0$ for $a > 1$.

# 6 A Stronger Charge Exponent and Optimal Annular Split

**Theorem 6.1** (Slow-charge range and improved summable bootstrap)**.** *Under the analytic inputs of Section 5, the first pooled-charge step of source Section 6 holds for every fixed $p_0 \in (0, 1/3)$, with a sufficiently large fixed $L_0$:*

$$\mathcal{R}(C_q(f)I - PD_q(f)P)\mathcal{R}^* \geq cS_{L_0,p_0}, \qquad C_q(f) = \frac{q+3}{3(q+1)} \sum_{j=0}^{q} f(j).$$

*It gives a weak integrated local comparison $C(q+1)^{1-p_0}(h+h')$. For any fixed $1 < p < 2$, choose a readout exponent $p < a < 2$. The centered electron-step budget for the summable weight $f_{L,p}$ is*

$$O(L^{-\theta}), \qquad \theta = \frac{p_0}{p+1}, \tag{11}$$

*by splitting an annulus $1 + m \asymp M$ at flux $T_M = C_0 M^{1+\theta}$. Constants can depend on $p, p_0$ and on all the retained analytic inputs. They are not asserted uniform as $p_0 \uparrow 1/3$.*

*Proof.* We first check every exponent restriction in the slow step. The exact electron scalar $\ell_q = C_{q-3}(f) + f(q) - C_q(f)$ is

$$\ell_q = \frac{\sum_{j=0}^{q-3}(j+1)(j+2)d_j}{(q+1)(q-2)} + \frac{2q}{3(q+1)}d_{q-2}, \quad d_j = f(j) - 2f(j+1) + f(j+2) \geq 0.$$

Expansion of coefficients proves this identity for every sequence $f$; convexity gives the sign. If $F(x) = x^{-1}\int_0^x f(t)\,dt$, then $\ell(x) = -2F'(x) + f'(x) = x^{-2}\int_0^x t^2 f''(t)\,dt$. Integration by parts gives $\int_k^\infty \ell(x)\,dx = 2F(k) - f(k)$. For $p_0 < 1$, $F(k) \leq f(k)/(1-p_0)$. Electron-step intervals of length three have disjoint interiors. The discrete-to-continuous error is bounded by $C(L+q)^{-2}$, whose tail, divided by $f(k)$, is at most $C/L$. Thus each flag pays at most $[(1+p_0)/(3(1-p_0)) + O(L^{-1})]f(k)$ over all electron ancestors. When two or more flags remain, sharing equally cuts this bound in half. The flag-creation credit is $[1/3 - O(L^{-1})]f(k)$. The net margin is

$$m(p_0) = \frac{1}{3} - \frac{1+p_0}{6(1-p_0)} = \frac{1-3p_0}{6(1-p_0)} > 0. \tag{12}$$

This is the restrictive upper endpoint $p_0 < 1/3$.

For clarity, the analytic errors also vanish on that whole range. Let $B_q = \int A_{q,p_0}\,dm$ and $J_q = \int A_{q,p_0}/v\,dm$. Splitting at $q = L$ and at the north/south halves gives

$$\sup_q B_q/f(q) = O(L^{-1}), \quad \sum_q J_q/f(q) = O((1+\log L)/L), \quad \sum_q \sup_m A_{q,p_0}(m)/f(q) = O(L^{-1}).$$

For the last estimate use $CL^{-2}$ below $L$, and $CL^{-p_0}q^{p_0-2}$ above $L$; $p_0 < 1$ suffices. For the one-sided error, the northern integral is bounded by

$$\frac{C}{(q+1)^2}\int_0^{q/2} \frac{f(m)(1+m)^{3/2}}{(L+1+m)^2}\,dm.$$

The southern contribution is at most $C\sqrt{q+1}f(q)/(L+q)^2$. Dividing both by $f(q)$ bounds the sum by $C\sqrt{q+1}/L^2$ below $L$ and by $Cq^{-3/2}$ above $L$, provided $p_0 < 1/2$. Their total is $O(L^{-1/2})$. This restriction is weaker than (12).

With one flag the sharing margin is unavailable. The filled-step calibration and the assumed one-hole comparison instead cost $C\sup_m A_{q,p_0} + CJ_q + C|\ell_q - \ell_{3\lfloor q/3\rfloor}| + Cf(q)(L+q)^{-2}$. The discrete

scalar differences divided by $f(q)$ sum to $O(L^{-1})$: below $L$ use $CL^{-2}$, above $L$ use $C(F(q)+f(q))q^{-2}$. All these costs vanish. The bounded terminal spaces have $D_q(f)-C_q(f)=-hI/3+O(L^{-1})h$. At $h=0$ use exact filled calibration, not an error bound. At a positive-hole node $S_{L,p_0}\geq hf(q)I$; each ancestor flux is visited at most once. Thus the stated summed errors are form inequalities on all coherent history superpositions. Choose $L_0$ so their total is below $m(p_0)/2$.

Schur averaging on the irreducible one-particle space gives exactly

$$\int D_q(f)^g\,dg=\frac{\sum_{j=0}^{q}f(j)}{q+1}\mathcal{N}.$$

The pooled estimate therefore bounds the rotation average of a state's flag weight by $h\sum_j f(j)/(3c(q+1))$. Multiplication by physical area $2\pi q$ and $\sum_j f_{L_0,p_0}(j)=O((q+1)^{1-p_0})$ gives the weak comparison asserted in the theorem.

For the second step set $a_M=(M/(L+M))^2$ and $f_M=f_{L,p}(M)$. The annular interaction has size at most $C(q+1)^{-2}f_Ma_M$. Split at $C_0M^{1+\delta}$, where $0<\delta<p_0/p$ for the moment. At moderate fluxes the weak comparison, divided by the descendant lower bound $hf(q)$, costs

$$Ca_MM^{p\delta}\sum_{q+1\geq M}(q+1)^{-1-p_0}\leq Ca_MM^{-(p_0-p\delta)}.$$

At larger fluxes the fixed-axis readout gives $W_m\leq CS_{L,p}/f_M$. Its annular area is $O(M)$. After centering by the common linear reference, the cost is

$$Ca_MM\sum_{q>T_M}(q+1)^{-2}S_{L,p}\leq Ca_MM^{-\delta}S_{L,p}.$$

Rapid shells do not invalidate this split: the source readout is uniform for every radius, including shells reaching the south; choose its tail order above the fixed radius exponent $B$. Calibration changes cost the already summed $J_q$ independently on selected annuli.

For $0<\eta<2$, $\sum_{M=1,2,4,\dots}a_MM^{-\eta}\leq C_\eta L^{-\eta}$, by summing $L^{-2}M^{2-\eta}$ below $L$ and $M^{-\eta}$ above it. The exponent delivered by this split is $\min\{\delta,p_0-p\delta\}$. Its maximum occurs when the two arguments are equal, namely $\delta=\theta=p_0/(p+1)$. All remaining common scalar and calibration errors are $O((1+\log L)/L)$ and hence smaller. This proves (11). □

**Corollary 6.2** (Concrete refinement)**.** *With $p_0=1/4$, the slow margin is $1/18$. With $p=3/2$, use $T_M=C_0M^{11/10}$ and obtain an $O(L^{-1/10})$ centered-electron budget. Keeping the source's $p_0=1/8$ but optimizing only the split already gives $O(L^{-1/20})$. The displayed source choice is $1/24$.*

These are improvements of an intermediate asymptotic error budget, not measured improvements of the final disorder threshold. Constants may increase, and the source provides no numerical locality constants. Letting $p_0\uparrow 1/3$ formally approaches exponent $2/15$ when $p=3/2$, but loses the slow margin. We claim every fixed choice below that endpoint, not the endpoint or a uniform limit in $p_0$.

## 7 Local Loss and a Quantified Relative Bound

The source constructs loss operators $J_y$ with particle loss between 2 and a fixed $L_{\mathrm{loss}}$, satisfying

$$\int J_y^*J_y\,dy=H,\qquad \int J_y^*HJ_y\,dy\leq H^2-cH,\qquad c>0. \tag{13}$$

These inequalities are uniform on Fock space, not just on the filled sector. Here is the source-specific reason retained blocks matter. After $B_0(g)$ removes a pair, a fixed neighborhood is partly emptied

using $K_{j,0} = 1 - n_j + t_j n_j$ and $K_{j,1} = \sqrt{1 - t_j^2} c_j$. Completeness is exact. On an even normal-ordered monomial the dual channel multiplies by $t_j$ for every occurrence of $c_j$ or $c_j^*$, giving modified rows with coefficients $t_i t_j v_p(i,j)$. Smoothness at the emptying cutoff and pair-tail estimates give error $\varepsilon_M = O(M^{-1/2}) + O(e^{-c'M})$ while preserving coefficient one on distant pair energy.

The rotationally averaged surviving pair tensor is $0 \le A_{q,M} \le I$. For any fixed deficit $\ell$, its coefficient satisfies

$$\lim_{q\to\infty} a_{q,M,\ell} = 1 - \frac{1}{2} \sum_{k=\ell}^{M-1} 2^{-k} \binom{k}{\ell}.$$

The infinite sum is 2, so this tends to zero as $M \to \infty$. Only a finite head is passed through the fixed-$q$ limit; no unbounded Clebsch–Gordan sum is interchanged without control. For fixed $\ell$, $\mathcal{L}_4(W_4 P_{q,\ell} W_4^*) \le C_\ell H$ by pair-generated highest-vector contraction and Schur averaging. Choose $M$ first to pay the 23 low blocks and $\varepsilon_M$, then $q$; the high blocks are paid by their retained coefficient one. This explains (13) and its quantifier order. In the source $c = 1/100$ can be chosen after these existential choices, but $L_{\mathrm{loss}} = 3M + 2$ and the locality constants remain unspecified.

### 7.1 A useful choice of chemical potential

The Lindblad dual is $\mathcal{D}(O) = \int J_y^* O J_y \, dy - \{H, O\}/2$. Equation (13) gives $h_t \le e^{-ct} h_0$ and expected particle loss at most $B h_0$, where $B = L_{\mathrm{loss}}/c$. At fixed $q$, compactness provides a zero-energy limit point; no zero mode has particle number above $N$. It follows sectorwise that $\mathcal{N} - N \le BH$.

**Proposition 7.1** (Quantitative coercivity)**.** *Fix $0 < \vartheta < 1$, choose $\mu = \vartheta/B$, and set $G = H + \mu(N - \mathcal{N})$. Then*

$$G \ge a(H + |N - \mathcal{N}|), \qquad a = \min\{1, \mu, (1-\vartheta)/(1+B)\}.$$

*Moreover $H \le uG$ with $u = (1-\vartheta)^{-1}$, and the loss evolution satisfies*

$$\langle G \rangle_t \le v \langle G \rangle_0, \quad v = (1-\vartheta)^{-1}, \qquad \langle N - \mathcal{N} \rangle_\infty \le (v/\mu) \langle G \rangle_0.$$

*Proof.* On $n \le N$, $G = H + \mu(N - n)$; on $n > N$, the excess bound gives $G \ge (1-\vartheta)H$ and $H + n - N \le (1+B)H$. These prove both coercivity and $H \le uG$. Energy decreases and lost number is at most $B h_0$, so $g_t \le g_0 + \mu B h_0 \le (1 + \vartheta u) g_0 = v g_0$. At a zero-energy limit point $g_\infty = \mu \langle N - \mathcal{N} \rangle_\infty$. □

The source uses a looser coercivity-based $G$-control constant; the direct estimate above avoids using $1/a$ in that step. A Fock gap for $G$ follows from coercivity, the filled-sector gap, and the integer particle-number separation. It is not a perturbed Fock gap assumption.

**Theorem 7.2** (Explicit relative-bound closure)**.** *Let the source's fixed polynomial locality norm of order $\ell$ define the class of Hermitian, neutral, physical $A_x$ with $A_x \Omega = 0$ and norm at most one per cell. Suppose:*

1. (13) *and Proposition 7.1 hold;*
2. *the zero-space charge bound for every such interaction $A = \sum_x A_x$ is $|\langle A \rangle_\rho| \le Z \langle N - \mathcal{N} \rangle_\rho$;*
3. *for $C_{xy} = [A_x, J_y]$, $D_{xy} = 1 + d(x,y)$, and some $2 < b < \ell - 2$, the volume constant is $\sup_y \sum_x D_{xy}^{-b} \le V_b$ and the integrated $D_{xy}^b C_{xy}^* C_{xy}$ interaction has locality norm at most $R_b$ per cell.*

*Set*

$$A_0 = Zv/\mu, \qquad A_1 = \frac{2}{c} \sqrt{V_b R_b u v}, \qquad K_* = \left( \frac{A_1 + \sqrt{A_1^2 + 4A_0}}{2} \right)^2.$$

*Then $-K_* G \le A \le K_* G$, uniformly in $q$. An interaction of size $S$ per cell costs $S K_* G$.*

*Proof.* At a fixed flux let $K_q$ be the best constant for this whole class. It is finite because each admissible interaction kills $\Omega$ on both sides and $G$ has a positive gap on $\Omega^\perp$. Since $J_y\Omega = 0$ follows from the first identity in (13), every $C_{xy}$ kills $\Omega$. Its square is positive, neutral, physical, and in the same fixed-order class. Assumption 3 therefore yields $\sum_x \int D_{xy}^b C_{xy}^* C_{xy}\, dy \le R_b K_q G$. The exact Lindblad identity and weighted Cauchy–Schwarz imply

$$\left|\frac{d}{dt}\langle A\rangle_t\right| = \left|\mathrm{Re}\sum_x \int \langle J_y^* C_{xy}\rangle_t dy\right| \le \sqrt{V_b R_b K_q h_t g_t}.$$

Use $h_t \le e^{-ct}h_0$, $h_0 \le u g_0$, $g_t \le v g_0$ and $\int_0^\infty e^{-ct/2}dt = 2/c$. The integrated cost is $A_1\sqrt{K_q}g_0$. At a zero-space limit point, Assumption 2 and Proposition 7.1 give $|\langle A\rangle_\infty| \le A_0 g_0$. Thus $K_q \le A_0 + A_1\sqrt{K_q}$. Solving the quadratic for $\sqrt{K_q}$ gives $K_q \le K_*$. Homogeneity gives the size-$S$ statement. □

The closure must use a fixed $\ell$, not an unspecified succession of higher norms. Source Section 3 proves $\|C_{xy}^* C_{xy}\|_{x,\ell} \le C D_{xy}^{-\ell}$, which leaves an integrable weight $D^{b-\ell}$ in dimension two. The condition $b > 2$ controls the other Cauchy–Schwarz factor. These are the actual locality requirements behind $R_b, V_b$.

## 8 Disorder Stability Along the Moving Ground Line

The coherent resolution, with $\kappa_q = (q+1)/(2\pi q)$, gives

$$\kappa_q \int |k_x\rangle\langle k_x|\, dA(x) = I, \qquad V = \kappa_q \int \varphi(x) c(k_x)^* c(k_x)\, dA(x).$$

Since CAR annihilator norms equal their one-particle norms, localized frame tails give local operator tails. Only $\|\varphi\|_\infty$ enters. Derivatives of a merely measurable symbol are never taken.

**Theorem 8.1** (Conditional explicit stability radius)**.** *Assume a simple filled zero line and gap $g_0 > 0$ for all $q$ in a stated range, Theorem 7.2, and the source's uniform propagation and time-filter estimates. On a bootstrap interval where the filled-sector gap of $H_s = H + sV$ exceeds $g_0/2$, suppose the transported, centered derivative interaction below has size at most $S$ per cell for $|s| \le s_0$. For any fixed $0 < \tau < 1/2$, the gap remains at least $(1-\tau)g_0$ throughout*

$$|s| \le \lambda_\tau = \min\{s_0, \tau/(SK_*)\}.$$

*If $S = 0$, interpret the second endpoint as infinity. The constants must be uniform over the allowed symbols and fluxes.*

*Proof.* Choose smooth odd real $F$ with $F(\omega) = -1/\omega$ for $|\omega| \ge g_0/2$ and $F = 0$ for $|\omega| \le g_0/4$. Its inverse Fourier kernel has all absolute time moments: derivatives of the frequency function are integrable at infinity, and its $1/\omega$ tail permits an integrable logarithmic singularity near time zero. Polynomial propagation therefore preserves locality. In an eigenbasis let $(Y_x)_{ab} = F(E_a - E_b)(V_x)_{ab}$. Then $Y_x^* = -Y_x$. With $Y = \int Y_x$, solve $U' = YU$, $U(0) = I$. Neutrality confines all matrix elements crossing the ground line to the $N$ sector. The usual finite-dimensional projection derivative is $P_s' = [Y, P_s]$, so $UP_0U^* = P_s$.

Define $Q_x = U^*(V_x + [H_s, Y_x])U$ and $A_x = Q_x - \langle\Omega, Q_x\Omega\rangle I$. For each excited $N$-sector eigenvector $a$, the ground-to-excited matrix element before conjugation is $(1 + (E_a - E_0)F(E_a - E_0))(V_x)_{a0} = 0$. Thus $A_x\Omega = 0$ *exactly*; subtracting an expectation alone would not ensure this. Hermiticity gives reverse vanishing. The assumed size bound and Theorem 7.2 give

$$-SK_*G \le \frac{d}{ds}\{U^* H_s U - E_0(s) I\} = \int A_x\, dA(x) \le SK_*G.$$

On the $N$ sector $G = H$. Integration in either direction yields $U^* H_s U - E_0(s) I \geq (1 - SK_*|s|)H$. Min–max gives the asserted gap on the bootstrap interval. At a first boundary point within $[-\lambda_\tau, \lambda_\tau]$, continuity of the ordered eigenvalues would give a gap at most $g_0/2$, contradicting $(1-\tau)g_0 > g_0/2$. The interval therefore extends to the stated endpoints, retaining simplicity. □

### 8.1 Exact obstructions and simple examples

For $H = \mathrm{diag}(0, g)$ and $A = \left(\begin{smallmatrix} 0 & v \\ v & 0 \end{smallmatrix}\right)$, $\langle \Omega, A\Omega \rangle = 0$ but $A\Omega \neq 0$ when $v \neq 0$. Testing $(1, t)$ with both signs of small $t$ shows that $|\langle \psi, A\psi \rangle| \leq C \langle \psi, H\psi \rangle$ fails for every finite $C$. This explains why exact spectral filtering is indispensable. The perturbed eigenvalue separation in this example is $\sqrt{g^2 + 4s^2v^2}$, and the ground vector rotates: a proof fixing the old ground vector would miss the actual mechanism.

For any real bounded symbol, let $a_\varphi = \inf_{c \in \mathbb{R}} \|\varphi - c\|_\infty = (\operatorname{ess\,sup} \varphi - \operatorname{ess\,inf} \varphi)/2$. On the $N$ sector a constant symbol contributes $cNI$. The stability theorem for normalized symbols therefore applies whenever $|s| a_\varphi \leq \lambda_\tau$; constant symbols preserve the gap exactly. By contrast the elementary norm bound $\|V - cN\| \leq N a_\varphi$ gives only $E_1 - E_0 \geq g_0 - 2|s| N a_\varphi$ by min–max. The distinction is between a proven uniform local estimate and an extensive norm estimate, not between two numerical diagonalizations.

Approximate annihilation requires a different conclusion. If only $\|A\Omega\| \leq \epsilon$, the off-diagonal ground-excited term can be bounded by $2\epsilon|\alpha|\|\psi_\perp\|$ and hence by $\delta \langle \psi_\perp, G\psi_\perp \rangle + \epsilon^2 |\alpha|^2/(\delta g)$ when $G \geq g P_\perp$. There is an additive ground-sector error. It cannot be silently absorbed into a multiple of $G$, which vanishes on the ground line.

## 9 Finite Verification, Benefits, and Limits

The author test program uses exact rational arithmetic on the source's seven integer rows, not rounded eigenvalue signs. It reconstructs the three-body margins and all 23 four-body $ZBZ$ matrices from the printed formulas. For a real symmetric matrix $M$, the recurrence $J_0 = I$, $c_b = \mathrm{Tr}(MJ_{b-1})/b$, $J_b = c_b I - MJ_{b-1}$ gives $\det(xI + M) = x^d + c_1 x^{d-1} + \cdots + c_d$. Nonnegative rational coefficients exclude a positive root, and real symmetry then excludes a negative eigenvalue. This proves only the specified finite PSD checks. The run record states the exact tested range and outcome, with no replacement of a continuum interval by samples.

Additional bounded tests check the pair-row normalization, the scalar second-difference identity, the allowance count 1222, the limiting three-body tail, the rational final margin, the optimized exponents, and the explicit two-level obstruction. Tests supplement the proofs; they do not supply unspecified locality constants, a uniform $q$ threshold, or a Lean kernel run. The source and private-read provenance includes literal paths and hashes. Cached upstream tree identity, when used, is identified as a dated captured comparison, not a fresh live blob audit.

### 9.1 Advantages and costs

The physical Gram argument retains valid certificates on redundant wedge ranges and avoids unstable numerical rank decisions. Its transfer error is measured in pair energy, so arbitrarily many spectator particles do not increase the coefficient. The charge refinement improves a genuine source error exponent and exposes the two restrictions $p_0 < 1/3$ and $p_0 < 1/2$ rather than fixing unexplained numerical choices. The relative-bound formula displays where loss rate, missing-particle cost, geometric summability, and centering strength enter the threshold.

The costs are substantial. Exact rational matrices develop large numerators; the local Fock factor has $2^{25}$ states if represented naively, although the proof avoids constructing it. Certified coupling enclosures and finite-flux three-body margins remain necessary for a numerical $q$ threshold. The

loss construction has $2^{3M}$ Kraus labels and constants depending on fixed $M$. Increasing the slow exponent can shrink its positive margin and worsen $L_0$. No runtime improvement, experimental disorder threshold, or historical novelty is demonstrated.

## 10 Future Extensions and Contribution Assessment

An effective threshold would require explicit bounds on pinning condition numbers, frame seminorms, propagation/filter moments, charge constants, and the finite-flux coupling errors $e_D(q)$. The displayed formulas identify where each bound is used but do not manufacture its value. Adaptive choices of $p_0, p, \delta$ should minimize the complete constants, not only maximize an exponent. Extending scalar Toeplitz disorder to other neutral local interactions requires verifying the same uniform interaction norms. Number-changing perturbations invalidate the fixed-sector argument and need a suitable perturbed grand-canonical gap. Degenerate ground spaces require block centering and control of intra-band splitting. Infinite-volume statements need a separate thermodynamic construction; an expanding finite sphere is not already a GNS theorem.

The realized assessment is C3/5, independently of L4 applicability: Theorem 4.1 gives an explicit source-specific physical error budget; Theorem 6.1 and Corollary 6.2 prove a stronger charge range and an optimized annular exponent; Proposition 7.1 and Theorems 7.2–8.1 give quantified restricted improvements. Classical quadratic-form, Schur, CAR, and spectral-flow mechanisms are credited to their sources. The report does not count the inherited global Laughlin gap or disorder stability theorem as a new Ram result.

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